

Abstract

Beyond the Game: Evaluating Environmental and Psychological Variables within Collegiate Athletics

KADE M ANDRUSKI

The focus of my research is how incorporating additional factors into sports data analysis can enhance the existing sports model. Traditional sports like football, basketball, baseball, soccer, and volleyball have long relied on basic statistics like scores, player performance, and game records to analyze outcomes. However, with the advancement of data collection and analytical technologies, it is now possible to take a deeper look at the various factors that influence game results. These factors include environmental conditions, psychological readiness, stadium features, and player-specific data like fatigue levels or academic stress. The purpose of this research is to explore how the integration of these additional elements into existing sports data can provide more accurate predictions for future games. This study investigates these influencing factors through a quantitative survey administered to 44 collegiate athletes across multiple sports and division levels, demonstrating their impact on sports statistics, and predictive accuracy. A mixed-methods analytical approach, including principal component analysis, k-means clustering, correlation analysis, and inferential statistics, was applied to the survey data to identify meaningful patterns and athlete profiles.

Table of Contents

Chapter 1: Introduction	7
Purpose of this Research	9
Research Questions	9
Significance of this Research.....	10
Chapter 2: Literature Review.....	11
History of Sport Statistics	11
Multifaceted Utilization of Statistics	12
The Role of Technology and Data Collection in Sports Analytics.....	13
Significant Physical Factors in Sports Performance	14
Significant Psychological Factors in Sports Performance	16
Integrating External Factors into Predictive Models	17
Comparative Approaches in Different Sports.....	17
Definition of Key Terms	19
Chapter 3: Methodology.....	20
Introduction.....	20
Research Methods and Methodology	20
Research Design	21
Data Collection	21
Survey Method	22

Open-Ended Response Analysis	22
Data Analysis	23
Quantitative Data Analysis:.....	23
Qualitative Data Analysis:	23
Ethical Considerations.....	24
Limitations and Future Research.....	24
Chapter 4: Findings	25
Participant Demographics.....	25
Descriptive Statistics	27
RQ1: Environmental Factors and Game Outcome Prediction.....	28
RQ2: Psychological Factors and Predictive Model Accuracy	30
RQ3: Integrating Athlete-level Contextual Variables into Predictive Models.....	31
Principal Component Analysis	32
K-Means Cluster Analysis.....	35
Correlation Analysis	38
Inferential Statistical Tests.....	41
Internal Consistency and Scale Reliability	42
Regression Analysis: Predictors of High-Stakes Stress	43
Open-Ended Response Themes	44
Summary of Findings	46

Chapter 5: Recommendations and Conclusions	48
Summary of Findings and Research Question Responses	48
Response to RQ1: Environmental Factors and Game Outcome Prediction	48
Response to RQ2: Psychological Factors and Predictive Model Accuracy	49
Response to RQ3: Integrating Contextual Variables into Predictive Models.....	50
Limitations.....	50
Future Research Directions	51
Conclusions.....	53
References.....	56

List of Tables

Table 1. Demographic Breakdown of Survey Participants.....	26
Table 2. Descriptive Statistics for Survey Scale Variables (N=44).....	27
Table 3. PCA Loadings — 4-Component Solution (N=42, 62.4% Variance Explained)	33
Table 4. K-Means Cluster Profile Means (k=2, N=42)	37
Table 5. Significant Pearson Correlations Among Survey Variables (N=44)	40
Table 6. OLS Regression: Predictors of High-Stakes Stress (N=42)	43

List of Figures

Figure 1. Greatest single external factor impacting performance as reported by survey respondents (n=44).....	29
Figure 2. Weather conditions identified as most disruptive to performance (multiple selections permitted; n=44).	30
Figure 3 Scree plot showing variance explained by each principal component. Four components were retained, explaining 62.4% of total variance.	35
Figure 4. Silhouette scores by number of clusters (k=2 to k=6). k=2 yielded the optimal solution (S=0.245).	38
Figure 5. Significant Pearson correlations among survey variables ($p < .05$). Bars represent r values; color intensity indicates correlation strength.	40
Figure 6. Themes identified in open-ended survey responses, by frequency of mention (n=30 respondents who provided substantive responses).....	46

Chapter 1: Introduction

Sports have always been a vital part of human society rooted deep in history. There are thousands of sports documented worldwide with each from different backgrounds and cultures consisting of various components, tactics, and strategy which come together to form competition. Over time, the sports from the past have undergone key changes and refinements to become the games we know today. As our understanding of sports evolves, so too does the field of sports analytics. Through the collection and analysis of many different statistical measures, this discipline utilizes data to procure insights into athletic performance and outcomes. In doing so this allows analysts to identify patterns which add to informed strategies and have the potential to offer competitive advantages. With many recent advancements in technology, the methods for analyzing sports data have become increasingly sophisticated; modern sports analytics leads to a richer understanding of the various factors that affect game results.

In college sports specifically, the stakes are high for athletes and institutions, so accurate predictions of a player's ability become a valuable tool. Leading collegiate teams across various sports consistently use statistical analysis data to enhance their preparation for their games. The use of predictive modeling in sports analytics has completely revolutionized how teams approach matches, enabling them to make decisions on empirical evidence rather than mere intuition. Yet, the effectiveness of these models hinges on recognizing a range of internal and external factors, such as environmental conditions, player psychology, and real-time dynamics of the game.

It is apparent to anyone that has watched a sporting event that a game is made up of more than just the numbers. Numerous external factors can have an impact on the outcome of the game. With each game there is a different venue which has its own unique characteristics such as lighting, altitude, and playing field conditions. The fan atmosphere can also impact the team's

morale, possibly giving the home team advantages. Beyond physical conditions, there are psychological factors to consider such as athlete mental space and team cohesion. Understanding the combination of these factors alongside traditional quantitative data is crucial to developing a better understanding of sports analytics as a whole.

The relevance of this research is underscored by the growing reliance on data analytics in the sports industry. Organizations seek competitive edges, gambling fans want better odds, and athletes strive to perform better, all of which strengthens the demand for a more precise predictive model. Accurate forecasts have the ability to significantly influence coaching strategies, player recruitment, and overall game tactics. Additionally, advancements in technologies, like machine learning and wearable data collection devices, offer undiscovered opportunities to refine predictive models by integrating a new spectrum of variables. Understanding the complexities of sports predictions is crucial not just for teams and coaches but also for fans engaged in the sport. As audiences become more data savvy, there is a growing expectation for potential player insights rooted in empirical analysis.

The aim of this thesis is to explore the factors that shape the accuracy of predictive models in college sports. By analyzing external variables such as weather, crowd behavior, and stadium characteristics, alongside internal elements like player psychology and health, the study seeks to offer a more comprehensive view of sports predictions. The goal is to create a framework that strengthens the reliability of predictive analytics, enabling teams to make better-informed decisions and elevate performance. In doing so, this research aims to add valuable insights to the sports analytics field, stressing the need to incorporate a wider array of factors when constructing predictive models.

Purpose of this Research

This research endeavors to explore the various factors impacting the accuracy of predictive models in college sports. By analyzing the interplay between environmental conditions, such as weather and stadium dynamics, and psychological factors, this study aims to provide a comprehensive understanding of the components that shape game outcomes. This investigation will directly address the outlined research questions, assessing how these multifaceted influences can be effectively woven into predictive frameworks. In doing so, it seeks to enhance the reliability of sports analytics, equipping coaches and teams to make more informed decisions that could ultimately lead to improved performance.

Ultimately, this research aspires to bridge the gap between theoretical frameworks and practical applications in sports analytics. By offering valuable insights into the various factors affecting predictive accuracy, it will not only enrich academic discourse but also serve as a practical resource for professionals in the sports industry. Through a thorough examination of the identified research questions, this study seeks to establish a foundation for future research endeavors, fostering continued exploration and innovation in the field of sports analytics.

Research Questions

RQ1: How do environmental factors, such as weather and crowd dynamics, affect the accuracy of game outcome predictions in college sports?

RQ2: In what ways does the psychological state of players, including stress and fatigue, influence predictive models for game results?

RQ3: How can athlete-level contextual variables, such as player fatigue and momentum shifts, be better integrated into sports prediction models?

Significance of this Research

This research holds vital importance for advancing the field of sports analytics, particularly within college athletics, where the pressure for accurate predictions is immense. As teams and coaches increasingly rely on data-driven insights to shape their strategies, a well-rounded idea on how outside factors affect their performance becomes critical. By investigating how environmental conditions, player psychology, and real-time dynamics impact predictive accuracy, this study seeks to establish a more comprehensive analytical framework on what factors truly affect the outcome of the game. The implications of this research extend beyond theoretical discourse; it aims to deliver actionable insights that could directly influence coaching decisions and team performance in real-world contexts.

Moreover, the significance of this research is magnified by the rising demand for transparency and sophistication in sports analytics. As fans and stakeholders engage more deeply with the data that drives sports outcomes, there is an expectation for analytical models to reflect the game's complexities. This study will bridge the gap between academic research and practical applications, fostering a deeper understanding of how various elements interact to shape game results. By providing a holistic view of predictive modeling that incorporates both quantitative data and qualitative factors, this research can enhance the overall experience for fans while contributing to a richer dialogue about the role of analytics in sports. The findings may inspire new methodologies and approaches to sports predictions, advancing how analysts and teams interpret data. In this way, the research not only enhances the current understanding of sports analytics but also paves the way for continued dialogue and development in the discipline, promoting a culture of continuous improvement in the prediction and analysis of athletic performance.

Chapter 2: Literature Review

This chapter aims to provide an overview of the different factors affecting the accuracy of predictive models in collegiate sports, focusing on environmental conditions and psychological aspects that influence game outcomes. It begins with a brief history of statistical analysis in sports and is followed by how we utilize the data today. Following is the analysis of what outside factors can alter the outcome of games. In essence, the fundamental goal of sports analytics has always been to gain a competitive advantage by understanding and interpreting data in ways that traditional observation cannot provide. By examining how these variables interact, the study seeks to improve the reliability of sports analytics.

History of Sport Statistics

The history of sports statistics dates back to the centuries of recorded sports. The first was statistics and easy metrics: how many goals were scored, how many points were accumulated, and how much distance was covered. These were basic performance indicators for coaches and analysts as a snapshot of individual and team success (Irwin et al., 2018). Yet, this band structure approach was usually too simple to reflect the subtleties of a game or an athlete's effect on the team. As sabermetrics rose in baseball in the 1970s on the shoulders of Bill James, so came the introduction of more sophisticated statistical techniques. In particular, James critiqued the conventions of baseball statistics, including batting average and the earned run average (ERA). He proposed more holistic metrics which better articulated a player's worth to the team (De Bosscher et al., 2016). For example, he also invented the On-Base Plus Slugging (OPS) statistic, a blend of on-base percentage and slugging percentage, to present a complete measure of a player's contribution to the offence. However, this was revolutionary, and the fruits of its labor ushered in other advanced metrics in the sports analytics world, from a Player Efficiency Rating

(PER) in basketball to Expected Goals (xG) in soccer and advanced pitching statistics like Fielding Independent Pitching (FIP) in baseball.

The introduction of predictive analytics came with the spread of statistical models across computational power. Statistical algorithms known as predictive modelling would be used to forecast future outcomes based on historical data and would become an inevitable tool for teams and analysts. The move from descriptive statistics to predictive analytics enabled performance, outcome, and strategy insights at a deeper level. The "Moneyball" revolution of using these advanced statistics in the management of a baseball team was indicative of the same, or how data-driven thinking could gain competitive advantage. Predictive analytics has since spread across sports and has affected not only player recruitment and development but also game tactics and in-game decisions (Teeter, 2021).

Multifaceted Utilization of Statistics

Sports statistics is no longer an essential metrics field but rather a sophisticated combination of traditional metrics and cutting-edge data science. Advanced statistics are now used in all sports to present a more detailed team and individual performance. PER and Win Shares are basketball stats meant to capture the value of a player that can't be neatly summed up in traditional box scores. Likewise, in soccer, Expected Goals (xG) has gained prominence as an influential metric to assess the ability of a team to score based on the scoring chances they have created, and, unlike scoring goals alone, can be considered a better indicator of a team's performance (Teeter, 2021). The application of machine learning algorithms has made big leaps towards predictive analytics in the sports arena. These algorithms can run through enormous amounts of historical data, and predict game outcomes, player performance, and throw out injury

risks. Predictive models are used by teams not only to predict individual players' success but also to assist in strategy for upcoming games. For example, predictive models could suggest the best lineups based on statistical analyses of past games using things such as each player's statistical output, opponents' flaws and the situation account (Shank & Lyberger, 2014).

Additionally, fan engagement with sports analytics has increased due to the rise in popularity of fantasy sports and sports betting. Now, the fans expect real-time data-driven insights about how the game is played, about how the players are doing as health and performance and statistical projections. SportsLine and ESPN platforms have published sites where common viewers and fans can access advanced statistics and predictive models, which are very common among coaches and analysts. Many of these platforms use proprietary models to predict pretty much anything related to outcomes, team performance, individual player statistics, etc (Karg & Funk, 2020). The democratization of data has also made fans' data savvier, and this has resulted in an increased expectation of fans to provide them with depth of sports complexities. However, there are enormous limitations in current models. Physical performance metrics are typically the coveted primary metrics used in predictive analytics, with what really affects the performance of an athlete being ignored: external environmental factors and the psychological aspects of an athlete's performance. In dynamic and unpredictable sporting environments, these missing variables can skew predictions and invalidate forecast outcomes.

The Role of Technology and Data Collection in Sports Analytics

However, the role of technology in data collection is ever evolving in predictive sports analytics. The introduction of real time data tracking systems, wearable devices, and advanced sensors has revolutionized the way that teams and analysts can now collect and put what they

collect to use. GPS trackers, heart rate monitors, and accelerometers are wearables which provide continuous feedback on an athlete's physical condition during practice and games (Shank & Lyberger, 2014). These tools enable coaches and analysts to track fatigue levels, muscle strain and recovery, revealing data that had never been able to be tracked before. Additionally, equipment, such as a basketball, a soccer ball, or a tennis racket, have sensors embedded within them, allowing the tracking of velocity, spin and trajectory and more so as further data for predictive models. Real-time data collection and analysis have enabled new ways to utilize sports prediction and make moves in response to live draft or event conditions. Through the usage of these technologies, predictions of these models can contain more granular and specific data for more precise forecasts and better-informed decision-making on both game planning and player management (Appelqvist et al., 2016).

Significant Physical Factors in Sports Performance

The environment in which a game is played, as a result, has been one of the primary external factors that affect athletic performance. Weather, including temperature, humidity, wind speed and precipitation, have long been understood to be important factors in sports outcomes. Further, according to Ely et al. (2017), marathon performance declines quite precipitously as Wet Bulb Globe Temperature (WBGT) rises, with slower runners degrading more than faster ones. Even in the present climate, elite athlete marathon performance was found to be approximately 1.7–4.5% slower in warmer conditions (WBGT 20–25°C), demonstrating the performance-limiting effects of temperature and humidity. Weather conditions also affect player performance in team sports, and this finding doesn't seem to be unique to marathon running. Wind and rain can alter the way a soccer ball, football or baseball bounces. High temperatures can aggravate fatigue during more prolonged endurance events.

Another physical factor that can affect performance to an equally profound extent is altitude. Athletes who train or compete at high altitudes have adapted to lower oxygen levels so that they perform better than those accustomed to sea-level conditions. But this physiological advantage is most obvious in endurance sports like cycling or long-distance running. In the NFL, for example, teams that play in high-altitude stadiums, such as the Denver Broncos, receive home-field advantage over teams visiting from a lower altitude, as the lower oxygen levels (at the higher altitude) cause visiting teams to adjust their game (De Bosscher et al., 2006).

Performance can also be affected by field conditions, such as the surface type on which the sport is played. This includes the sorts of things like injury rates amongst players appearing to depend on playing on grass vs playing on artificial turf. For example, grass surfaces provide more traction and are therefore more useful for soccer, football, etc., whilst artificial turf (which is harder and less forgiving) can make injury more common (Ely et al., 2007). Modelling these environmental variables in conjunction with conventional performance metrics allows a more accurate prediction of outcomes in sports where there are important external field conditions to consider.

Additionally, many physical factors can affect athlete performance: travel and jet lag. Effects of disruption to circadian rhythms are known from teams that travel across multiple time zones. Recent research indicates that teams who travel in an eastward direction (e.g., from the U.S. to Europe) suffer disproportionately from ergonomics because of problems with sleeping among team members and decreased recovery times (De Bosscher et al., 2006). Understanding these travel-related variables is especially critical for collegiate athletes who may have a series of compact schedules that involve numerous games during a season, with long-distance travel to the games adding up fatigue throughout the season.

Significant Psychological Factors in Sports Performance

Psychological elements other than physical were found to play a significant role in athletic performance. For example, home-field advantage is mostly the result of mind and social reality. When a crowd of familiar fans support athletes from a home stadium, as familiarity increases the emotional boost athletes receive from fans that can motivate and focus on their performance, athletes have the comfort and familiarity of home giving them something to play and focus on performance. Still, teams visiting a place might need to be more focused by being away (Appelqvist et al., 2016).

Psychological factors which do affect performance include stress and mental health. But high-stakes games like playoffs and championship games put such pressure on athletes. Stress from this pressure can decrease concentration, decision making and physical performance. For example, anxiety could result in poor execution of skills (Fradejas Medrano & Espada Mateos, 2018). At the same time, also mental fatigue will result in the athlete's inability to maintain optimal performance levels throughout a game and so on. Psychological stress measures, i.e. levels of cortisol or heart rate variability, make for valuable data about the psychological conditions of athletes, which a predictive model should model better than physical parameters.

Another psychological factor impacting performance is winning and losing streaks. Teams that are currently on a winning streak will psychologically boost themselves, providing players with increased confidence and unity. Alternatively, teams losing a few in a row may see a matter of morale and exacerbate their performance problems. According to the research, the outcomes of games are influenced by momentum, not just the good kind and the bad kind. Finally, predictive models that incorporate these psychological elements give a better idea of how teams will act by their present mental state (Bali, 2015).

Student athletic life presents some unique psychological challenges. Athletes in college are constantly being pulled at by academics, athletics and social life, leading to burnout and stress. Athletes are in a fight to see if they can balance their physical and emotional needs, which can put an immense amount of stress on the athlete. Since these aspects are not incorporated in the predictive models of predictive models, they can improve the accuracy of predicting collegiate sports on our true complexity (De Bosscher et al., 2016).

Integrating External Factors into Predictive Models

Despite the increasing sophistication of sports analytics, current predictive models often include environmental and psychological variables. The difficulty in quantifying and integrating such factors in traditional models has led to this to a large extent. But, with the introduction of real-time data tracking technologies and machine learning algorithms, there is enormous potential to incorporate these external and psychological factors into more encompassing models. For instance, wearables can collect physiological data like heart rate, body temperature and movement patterns, as well as environmental data, real-time sample, weather conditions. Parameters extracted from these data sources can be combined to offer more accurate forecasts utilized by predictive models, which account for the interplay of complex real-world physical, environmental and psychological parameters.

Comparative Approaches in Different Sports

The core principles of predictive analytics don't change by sport, but the precise ways of modelling and measuring can. For example, in the sport of baseball, common uses for predictive models include identifying positives or negatives on an individual performance level for batting average, home runs, earned run averages and others, such as Wins Above Replacement (WAR).

However, as the sport is inherently based on individual outcomes, these models are extremely focused on individual outcomes and are often easy to isolate individual performance. On the contrary, sports such as soccer, football, and basketball require models that involve team interaction since these games are more team-based. An example of such as team-centric model would be soccer's Expected Goals (xG) metric, which predicts the probability that a goal will be scored based on how likely a shot is given and measured in shot location, angle, and defense positioning (Jauhiainen, 2023). [In basketball], another model to which you have probably been exposed is the Box Plus-Minus (BPM) which seeks to quantify a player's impact on the team—offensively and defensively. To summarize, different ways different sports are using predictive models will illustrate the application of analytics to different domains, and how context plays a role in developing models. These differences also indicate that for certain factors, for example, environmental conditions or team dynamics, there can be more or less impact in various sports, representing, in turn, new possibilities for cross-sports research.

Definition of Key Terms

Environmental Factors – as related to sports, external factors that can impact an athlete's performance such as temperature, humidity, pollution, playing surface, etc.

Psychological Factors – internal factors that can impact an athlete's performance and well-being such as motivation, anxiety, confidence, or team cohesion

Home-Field Advantage – the tendency for athletes to perform better when competing at their home facility

Team Dynamics – refers to the relationships and interactions between team members which as the capability to affect performance

Mental Fatigue – a state of tiredness that occurs when the brains energy levels are depleted which can be induced by many factors

Travel Fatigue – the feeling of exhaustion or tiredness that can be the result of extensive traveling

Field Conditions – the state of the designated playing area or surface. Many factors go into this such as court/ field material, lighting, and overall ambiance

Thematic Analysis – a qualitative research method that deals with examining data to identify or report any themes in the data

Correlation Analysis – a statistical method used to determine the strength or direction of the relationship between variables

Performance Metrics – quantifiable measurement used to assess athletes' performance such as speed, strength, accuracy, etc.

Chapter 3: Methodology

Introduction

This chapter outlines the methodology employed in this study, which investigates how environmental and psychological factors affect the performance of college student-athletes. Weather conditions, field type, psychological stress, team dynamics, and other situational variables discussed in the literature review were examined. To achieve this purpose, this research sought to understand more deeply the relationship between external and internal conditions and athletic performance, with the goal of informing coaching strategies and training regimens. The study employs a quantitative survey methodology, collecting data from 44 collegiate athletes across multiple sports and division levels. Data analysis techniques including descriptive statistics, correlation analysis, principal component analysis, k-means clustering, and inferential tests were applied to arrive at verifiable conclusions.

Research Methods and Methodology

A quantitative survey methodology was employed as the primary research method for this study, followed by open-ended qualitative responses. This approach was selected because it enables broad, generalizable data collection across a diverse athlete population while retaining the flexibility to capture nuanced experiential insights.

Quantitative research methods were used to collect broad, generalizable data about college athletes' experiences and the mechanisms through which those experiences affect performance. The survey generated numerical data that, when analyzed statistically, revealed meaningful patterns and trends.

Qualitative research methods approaches were used to capture qualitative themes from open-ended survey responses. These responses provided richer insight into factors that predetermined athlete performance beyond what structured Likert-scale items could measure.

Research Design

The research design was adopted as exploratory. An exploratory design was chosen because it had not been sufficiently studied with what combination of environmental and psychological factors impact college athlete performance. It will bring new insights and create a basis for future research in this field. The study aims to be descriptive in an attempt to explain the experiences and perceptions of college athletes as to specific environmental and psychological variables. The study is correlational in design and essentially studied relationships between environmental and psychological variables and performance outcomes. The study examines patterns of responses across athletes from different sports to relate changes in performance with some external factors. This exploratory research design allows for flexibility in how findings are interpreted as new insights emerge from the data. The design is grounded in the literature reviewed in Chapter 2 and performed through the survey instrument described in the Data Collection section.

Data Collection

Data was collected through a structured online survey distributed to collegiate athletes in spring 2026. The survey instrument captured both quantitative ratings and open-ended responses to provide a comprehensive picture of athlete perceptions.

Survey Method

In this study, a survey with a wide selection of college student-athletes among different teams and schools in the spring 2026 semester was the primary data collection tool. The survey aims to collect quantitative data about athletes' perceptions of the influences of environmental as well as psychological factors on their performance. Likert scale and other closed-ended questions on the significance of weather, field conditions, travel, stress, motivation, mental fatigue, team dynamics, and home-field advantage will be involved. Demographic data such as sport, age, gender, and competition level was collected to account for potential variation. The means of survey dispersion amongst university sports programs and social media aims to gather a diverse sample of both team and individual sports. Informed consent documents were constructed and provided to participants outlining the study's purpose, how confidentiality was being handled, and that participants had a right to withdraw from the study at any time. This should give a wide and varied view of the factors that affect athletic performance.

Open-Ended Response Analysis

The survey included one open-ended question inviting respondents to identify external or psychological factors they believed were underrepresented in sports analytics. Thirty of 44 participants (68.2%) provided substantive responses. These responses were analyzed using thematic analysis — an inductive coding process in which recurring ideas were identified, grouped, and refined into nine coherent themes. This qualitative component complements the quantitative survey data by surfacing factors that Likert-scale items alone could not capture, including coach-athlete relationship quality, family stressors, and in-game emotional momentum that would provide rich information to answer the research questions.

Data Analysis

Data analysis methods incorporate quantitative and qualitative methods to address all aspects of the research questions.

Quantitative Data Analysis:

Descriptive statistics are used to analyze the survey data and quantify the response with means and standard deviations for each variable. With this, it provides the general trends of how athletes understand how environmental and psychological factors influenced their performance. The potential relationships between such factors as perceived stress level, weather conditions, and performance outcome are explored by correlation analysis. For instance, the study aims to analyze whether athletes who said they had higher stress levels scored lower on performance. These analyses were performed using Python (scipy, scikit-learn, numpy) to identify statistically significant patterns in the data.

Qualitative Data Analysis:

Thematic analysis was applied to the open-ended survey responses to identify recurring themes in athletes' perceptions of underrepresented performance factors. The process began with open coding of responses, after which initial themes were grouped and refined. Nine thematic categories emerged, with coach-athlete relationship quality and family or personal life stressors representing the most frequently cited. These themes are reported alongside the quantitative findings in Chapter 4 extreme heat made them run faster or slower and how team dynamics impinge their psyches while competing. The analysis will start with an open coding process, where initial identity themes will be identified and grouped. These themes then will be refined and categorized as the analysis continues, providing a deeper understanding of the athletes'

experiences. The reported experience data should also be analyzed to see if there were any correlations with actual performance outcome statistics; for instance, those who report high stress levels might have lower performance statistics than those who report low stress levels.

Ethical Considerations

During the research process, ethical guidelines will be applied to the study. All participants were informed about the study, how they will be approached, what information will be collected, and how anonymity of such responses will be ensured. Participants were informed at any time that they can withdraw from the study without penalty. To protect participants' identities, all personal data is anonymized. This was reviewed and approved by the university's ethical review board for ethical standards.

Limitations and Future Research

The primary limitations of this study include the sample size of $N=44$, which constrains statistical power for inferential analyses and the overrepresentation of volleyball athletes (47.7%), which may not reflect the full breadth of collegiate sport experience. All data are self-reported, introducing the possibility of recall bias and social desirability effects. These limitations are discussed in detail in Chapter 5, alongside recommendations for future research that addresses them.

Chapter 4: Findings

This chapter presents the findings from the quantitative survey administered to 44 collegiate athletes in spring 2026. The survey collected data on athletes' perceptions of environmental and psychological factors affecting performance, as well as their engagement with sports analytics. Data was analyzed using descriptive statistics, Pearson correlation analysis, principal component analysis (PCA), k-means clustering, independent samples t-tests, Mann-Whitney U tests, one-way ANOVA, chi-square tests, Cronbach's alpha reliability analysis, and ordinary least squares regression. All analyses were conducted using Python (scikit-learn, scipy, numpy). Findings are organized by research questions and presented alongside supporting tables and figures.

Participant Demographics

Table 1 presents the demographic characteristics of the 44 survey respondents. The sample was predominantly NCAA Division II (n=18, 40.9%) and Division I (n=11, 25.0%), with additional representation from NAIA (n=6, 13.6%) and NCAA Division III (n=5, 11.4%). Volleyball was the most represented sport (n=21, 47.7%), followed by soccer (n=6, 13.6%), baseball/softball (n=5, 11.4%), and basketball, football, and boxing (n=3 each, 6.8%). The gender distribution was 55% male (n=24) and 43% female (n=19). Ages ranged from 18 to 29, with the majority between 19 and 24. Fifteen respondents (34.1%) were graduates or no longer enrolled, while 8 (18.2%) were graduate students, indicating that many participants had substantial competitive experience.

Table 1. Demographic Breakdown of Survey Participants

Characteristic	Category	n	%
Sport	Volleyball	21	47.7%
	Soccer	6	13.6%
	Baseball/Softball	5	11.4%
	Basketball / Football / Boxing	3 each	6.8% each
	Track & Field	2	4.5%
Division	NCAA Division II	18	40.9%
	NCAA Division I	11	25.0%
	NAIA	6	13.6%
	NCAA Division III	5	11.4%
	Club/Amateur	3	6.8%
Gender	Male	24	54.5%
	Female	19	43.2%
Academic Year	Graduated/No longer enrolled	15	34.1%

Characteristic	Category	n	%
	Graduate Student	8	18.2%
	Senior	7	15.9%
	Freshman	6	13.6%
	Junior / Sophomore	7	15.9%

Descriptive Statistics

Table 2 presents descriptive statistics for all Likert-scale and ordinal variables included in the survey. Team cohesion and communication was rated as the most important factor ($M=4.64$, $SD=0.72$), followed by game-day routine impact ($M=3.41$, $SD=1.01$) and crowd size and noise ($M=3.32$, $SD=1.10$). Physical environmental variables, weather ($M=2.59$, $SD=1.25$), playing surface ($M=2.64$, $SD=1.04$), and travel/time zone effects ($M=2.73$, $SD=0.90$), received moderate ratings. Academic deadline conflict was rated at $M=3.07$ ($SD=1.09$), indicating that academic demands interfere with game preparation on a regular basis for a notable portion of the sample.

Table 2. Descriptive Statistics for Survey Scale Variables (N=44)

Variable	N	Mean	SD	Min	Max
Team cohesion & communication	44	4.64	0.72	2	5

Variable	N	Mean	SD	Min	Max
Game-day routine impact	39	3.41	1.01	1	5
Crowd size & noise impact	44	3.32	1.10	1	5
Academic deadline conflict	44	3.07	1.09	1	5
Lighting conditions impact	22	3.00	0.71	1	4
Travel / time zone impact	44	2.73	0.90	1	5
Playing surface impact	44	2.64	1.04	1	5
Weather conditions impact	44	2.59	1.25	1	5
High-stakes stress (ordinal)	44	2.89	0.84	1	4
Travel fatigue (ordinal)	42	2.81	0.79	1	4
Home crowd advantage (ordinal)	44	3.30	0.71	1	4

RQ1: Environmental Factors and Game Outcome Prediction

The first research question investigated how environmental factors — including weather, playing surface, crowd dynamics, and travel — affect the accuracy of game outcome predictions in college sports. Survey responses reveal that travel fatigue was identified as the single greatest external factor impacting performance by the largest share of athletes (n=15, 34.1%), narrowly surpassing weather conditions (n=13, 29.5%) and home/away game location (n=10, 22.7%). This

finding is notable given that the existing predictive modeling literature emphasizes weather far more prominently than travel logistics.

Among weather conditions, cold temperatures were most frequently cited as disruptive ($n=23$), followed by heat ($n=12$), wind ($n=12$), and humidity ($n=11$). Sixty-one percent of athletes reported that home crowd support provides at least a slight performance advantage, with 45.5% describing it as a strong advantage, consistent with the home-field advantage literature (Appelqvist et al., 2016). Travel fatigue was reported as affecting away game performance at least sometimes by 59.1% of respondents ($n=26$), with 13.6% indicating it always affects their performance.

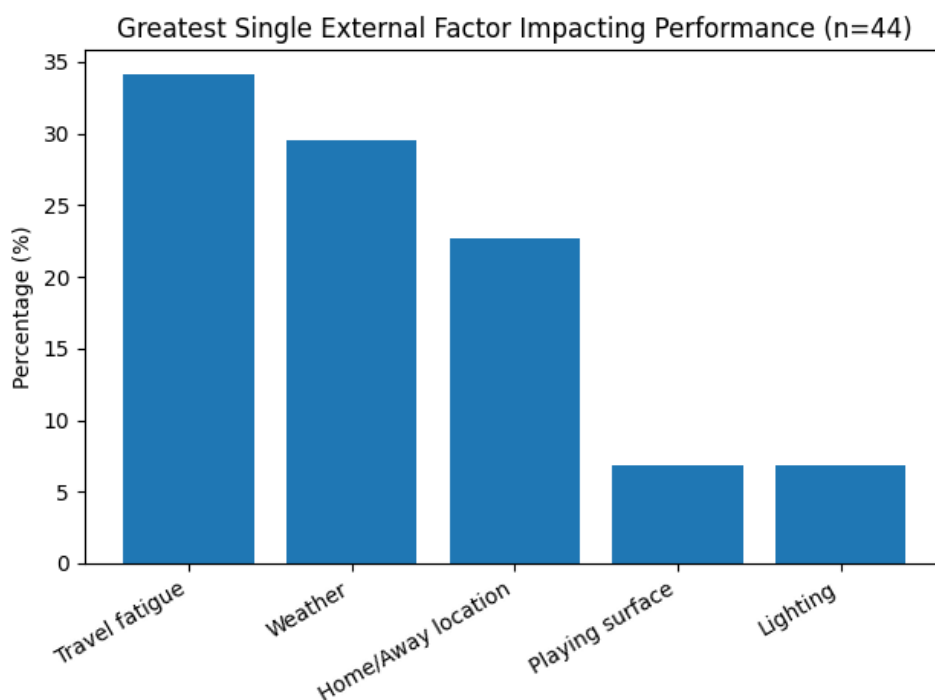


Figure 1. Greatest single external factor impacting performance as reported by survey respondents ($n=44$).

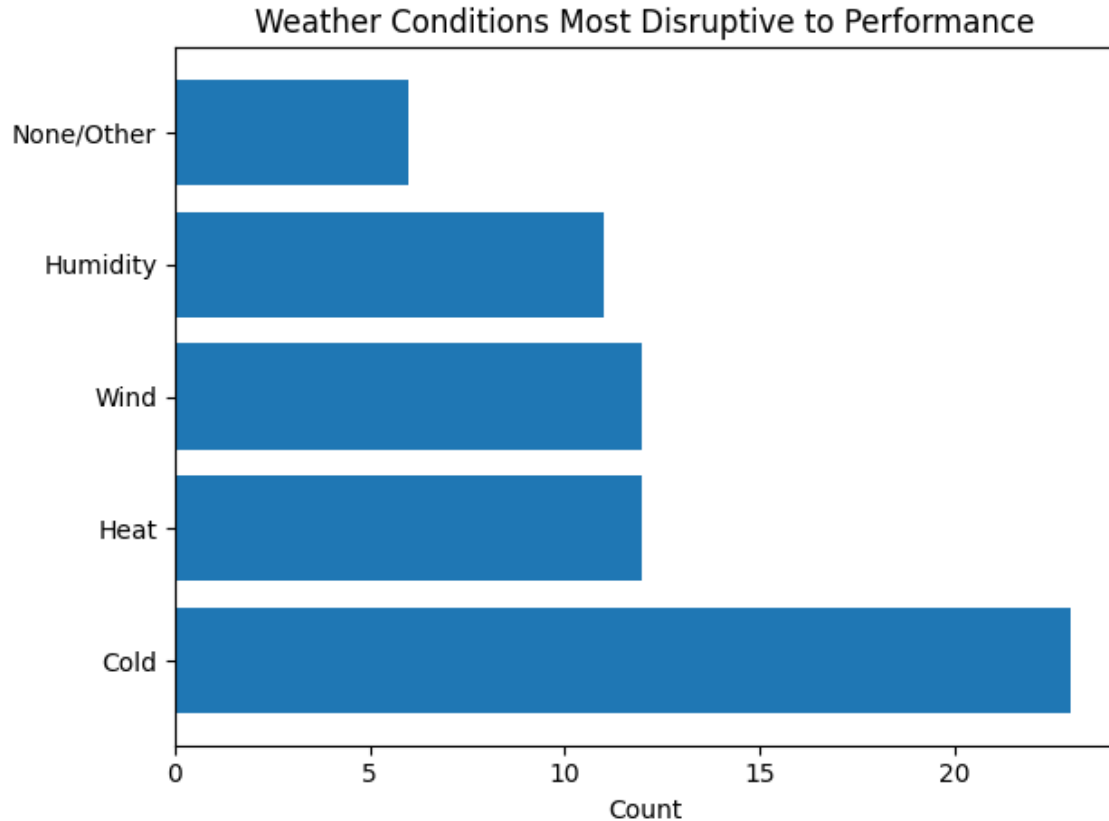


Figure 2. Weather conditions identified as most disruptive to performance (multiple selections permitted; n=44).

RQ2: Psychological Factors and Predictive Model Accuracy

The second research question examined how the psychological state of players influences predictive models for game results. Team cohesion emerged as the dominant psychological factor, receiving the highest mean rating in the entire survey ($M=4.64$, $SD=0.72$), with 81.8% of athletes rating it a 5 out of 5. This finding strongly supports the theoretical framework advanced in Chapter 2, particularly the work of Fradejas Medrano and Espada Mateos (2018), who identified team cohesion as a primary predictor of collective performance outcomes.

High-stakes game stress was reported by 72.7% of athletes, with 25.0% indicating it significantly affects their performance. Only 9.1% reported performing better under pressure. Regarding losing streaks, 43.2% of athletes reported that consecutive losses motivate them to work harder, while 40.9% reported slight decreases in motivation and confidence, a near-even split that underscores the heterogeneity of psychological responses to adversity.

A particularly noteworthy finding concerns the academic-athletic interface. When asked whether mental fatigue from schoolwork had ever affected their in-game performance, 84.1% of respondents answered affirmatively (57.0% a few times; 27.3% multiple times), with only 15.9% reporting it never occurred. Academic deadline conflict with game preparation was rated at $M=3.07$ ($SD=1.09$), with the modal response being 3 (moderate conflict). This academic fatigue dimension is absent from virtually all existing predictive frameworks for collegiate sports performance and represents a novel contribution of the present research.

RQ3: Integrating Athlete-level Contextual Variables into Predictive Models

The third research question addressed how athlete-level contextual variables can be better integrated into sports prediction models. Athletes' own perceptions of statistical prediction are highly informative on this point: 56.8% of respondents stated that statistics cannot fully predict game-day performance, compared to only 38.6% who considered statistics somewhat accurate and 2.3% who considered them highly accurate. This near-majority skepticism from the athletes themselves constitutes direct empirical support for the central thesis that current analytics models are incomplete.

Regarding analytics engagement, 45.5% of athletes review their own game film after every game, with an additional 40.9% doing so occasionally. However, coaching staff analytics

use was more limited: only 34.1% of athletes reported their coaching staff uses analytics regularly, while 50.0% reported occasional use and 15.9% reported rare use. This gap between athlete self-reporting behavior and institutional analytics adoption represents a practical barrier to the comprehensive predictive frameworks proposed in this thesis.

Principal Component Analysis

A principal component analysis (PCA) was conducted on 13 survey variables using a standardized input matrix (N=42 after listwise deletion). The scree plot and cumulative variance criterion indicated that a four-component solution was appropriate, collectively explaining 62.4% of total variance. Table 3 presents the component loadings for the retained solution.

PC1, labeled Pressure-Sensitivity (25.9% variance), captured broad external stressor sensitivity, with primary loadings on crowd impact (0.403), travel impact (0.381), travel fatigue (0.388), academic conflict (0.366), and home advantage (0.303). This component suggests that athletes do not neatly distinguish between environmental and psychological stressors — they experience them as a unified burden. PC2, labeled Physical Environment Sensitivity (13.8%), was driven by weather impact (0.427), playing surface (0.496), and losing streak response (0.520), capturing athletes who are particularly attuned to physical and conditions-based disruption. PC3, labeled Analytics Engagement (12.8%), was defined by film review frequency (0.651) and coaching analytics use (0.493), identifying a data-engagement dimension entirely independent of stress or environmental sensitivity. PC4, labeled Competitive Stress Response (9.9%), was dominated by high-stakes stress (0.698), negatively separated from team cohesion (-0.269), capturing acute psychological pressure under competitive conditions.

The emergence of a distinct Analytics Engagement component (PC3) is theoretically significant. It implies that athletes' and coaches' openness to data-driven approaches is a latent behavioral disposition, not a byproduct of their stress levels or environmental sensitivity. Interventions designed to increase analytics adoption must therefore target this dimension directly.

Table 3. PCA Loadings — 4-Component Solution (N=42, 62.4% Variance Explained)

Variable	PC1 Pressure-Sensitivity	PC2 Physical Env.	PC3 Analytics	PC4 Comp. Stress
Weather impact	0.120	0.427*	-0.281	-0.286
Playing surface impact	0.154	0.496*	0.192	-0.009
Crowd impact	0.403*	0.036	-0.169	-0.125
Travel/timezone impact	0.381*	0.057	0.172	-0.251
Academic conflict	0.366*	0.037	-0.305	-0.058
Team cohesion	0.201	-0.290	-0.157	-0.269
High-stakes stress	0.170	-0.016	0.174	0.698*
Travel fatigue	0.388*	-0.243	-0.080	0.035
Home advantage	0.303	-0.356	-0.021	0.102

Variable	PC1 Pressure-Sensitivity	PC2 Physical Env.	PC3 Analytics	PC4 Comp. Stress
Losing streak response	0.249	0.520*	0.014	0.214
Film review frequency	0.142	0.025	0.651*	-0.091
Coaching analytics use	0.165	-0.134	0.493*	-0.316
Academic fatigue in-game	0.316	-0.078	-0.062	0.334
% Variance explained	25.89%	13.77%	12.81%	9.93%

*|Loading| ≥ 0.35 , indicating primary contributor to component.

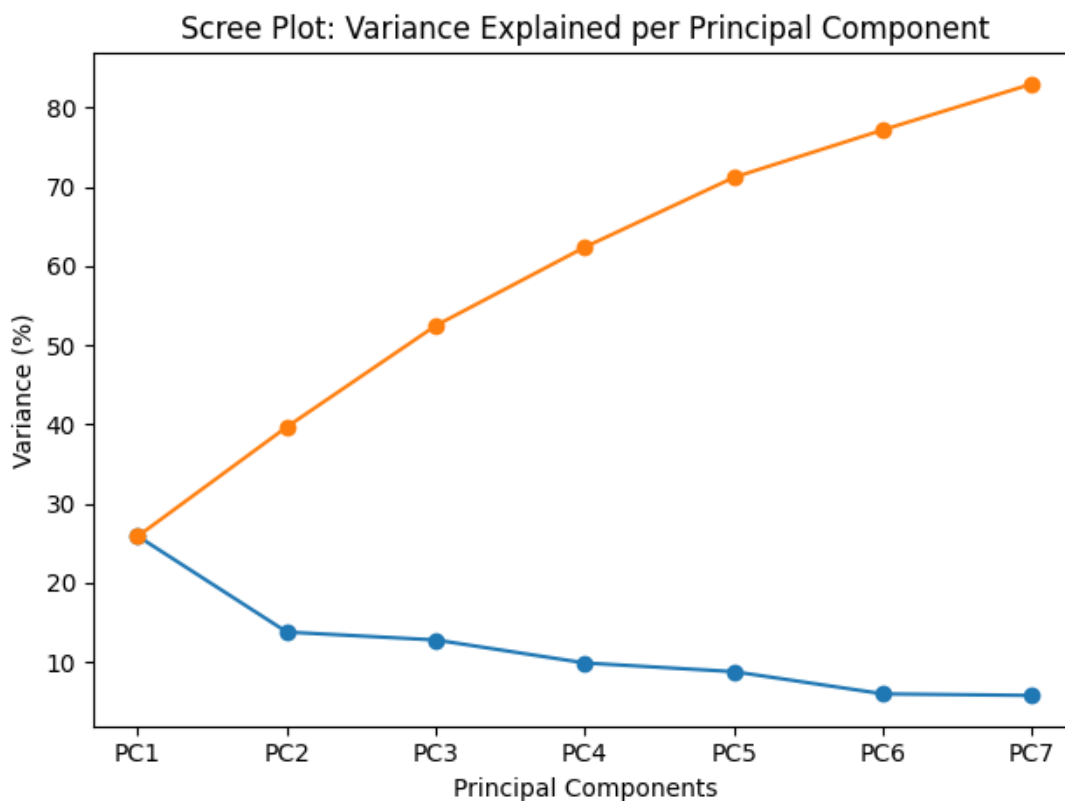


Figure 3 Scree plot showing variance explained by each principal component. Four components were retained, explaining 62.4% of total variance.

K-Means Cluster Analysis

K-means cluster analysis was performed on the four principal component scores to identify meaningful athlete profiles. Silhouette coefficients were computed for $k=2$ through $k=6$. The two-cluster solution yielded the highest silhouette score ($S=0.245$), indicating $k=2$ as the optimal partition. Although silhouette values below 0.50 suggest moderate rather than strong cluster separation, given the small sample ($N=42$) and the inherent heterogeneity of self-reported survey data, the two clusters produced theoretically coherent and meaningfully differentiated athlete profiles.

Cluster 0 (n=19, 45.2%), labeled High-Sensitivity Athletes, scored consistently higher across all environmental and psychological dimensions. Notably, this cluster reported substantially higher travel impact (M=3.26 vs. 2.26, Delta=1.00), crowd impact (M=3.74 vs. 2.96, Delta=0.78), and academic fatigue in-game (M=1.47 vs. 0.83, Delta=0.64). Cluster 1 (n=23, 54.8%), labeled Low-Sensitivity Athletes, showed relative insularity from external stressors across all measured dimensions. Table 4 presents full cluster mean comparisons.

The practical implication of this two-profile solution is substantial. If nearly half of collegiate athletes are systematically more sensitive to travel, crowd, and academic stressors, then a single predictive model applied uniformly across a roster will systematically underpredict performance disruption for those athletes. Athlete cluster membership, determinable via a brief survey, could serve as a categorical feature input in next-generation predictive frameworks.

Table 4. K-Means Cluster Profile Means (k=2, N=42)

Variable	Cluster 0 High-Sensitivity (n=19)	Cluster 1 Low-Sensitivity (n=23)	Delta
Weather impact	2.84	2.44	+0.40
Playing surface impact	2.95	2.35	+0.60
Crowd impact	3.74	2.96	+0.78
Travel/timezone impact	3.26	2.26	+1.00
Academic deadline conflict	3.37	2.78	+0.59
Team cohesion	4.89	4.43	+0.46
Coaching analytics use	2.42	2.00	+0.42
Academic fatigue in-game	1.47	0.83	+0.64
Silhouette score	—	—	S=0.245



Figure 4. Silhouette scores by number of clusters ($k=2$ to $k=6$). $k=2$ yielded the optimal solution ($S=0.245$).

Correlation Analysis

Pearson correlation coefficients were computed for all 12 primary scale variables. Table 5 presents statistically significant correlations ($p<.05$). Twelve of 66 possible pairs reached significance. The strongest correlation was between perceived travel impact and travel fatigue ($r=0.545$, $p<.001$), demonstrating strong convergent validity between the two travel-related measures and suggesting they tap a common underlying construct of travel-induced performance disruption. Travel fatigue also correlated significantly with academic fatigue in-game ($r=0.492$, $p=.001$), indicating that physically and mentally depleted athletes, burdened by both road travel

and academic pressures, represent a compounded vulnerability not captured by any single measure in isolation.

Film review frequency and coaching analytics use were significantly correlated ($r=0.468$, $p=.002$), suggesting that analytics culture is self-reinforcing: programs in which coaching staff regularly employ data tend to produce athletes who also engage in individual film review. Crowd impact demonstrated the broadest pattern of associations, correlating significantly with team cohesion ($r=0.459$, $p=.002$), academic conflict ($r=0.455$, $p=.003$), travel impact ($r=0.443$, $p=.003$), travel fatigue ($r=0.361$, $p=.019$), and home advantage ($r=0.384$, $p=.012$). This pattern reinforces the PC1 Pressure-Sensitivity component, crowd sensitivity co-varies with a broad constellation of external stressor perceptions.

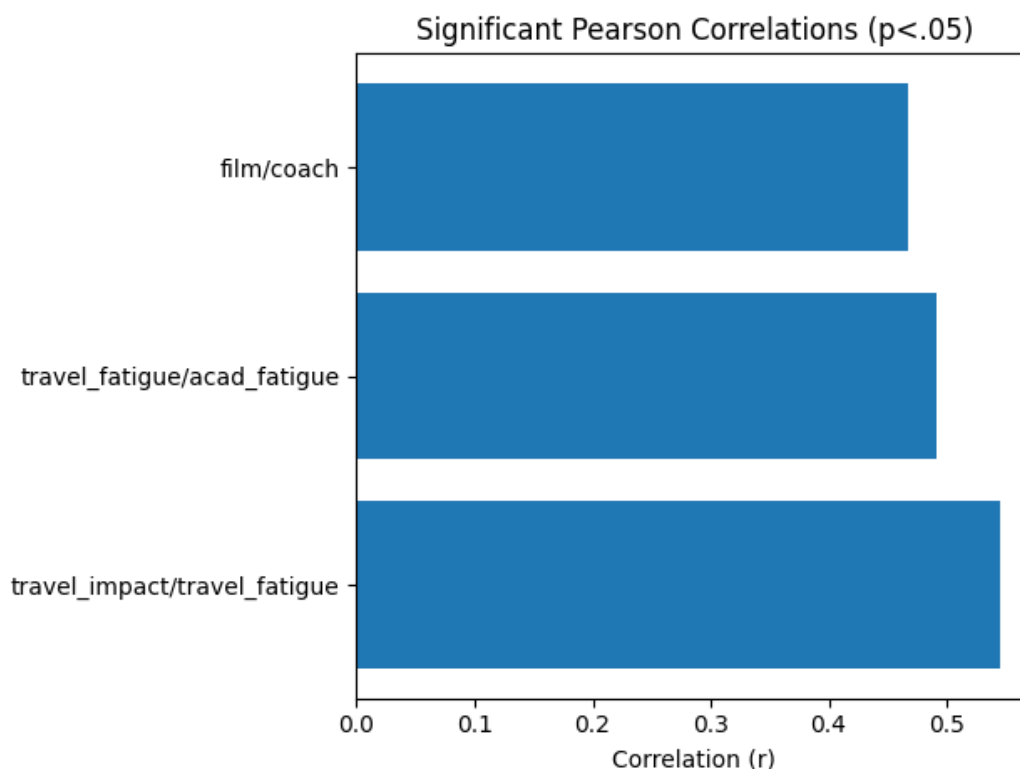


Figure 5. Significant Pearson correlations among survey variables ($p < .05$). Bars represent r values; color intensity indicates correlation strength.

Table 5. Significant Pearson Correlations Among Survey Variables (N=44)

Variable Pair	r	p-value	Significance
Travel impact ↔ Travel fatigue	0.545	.0002	***
Travel fatigue ↔ Academic fatigue	0.492	.001	***
Film review ↔ Coaching analytics use	0.468	.002	**
Travel fatigue ↔ Home advantage	0.472	.002	**
Crowd impact ↔ Team cohesion	0.459	.002	**
Crowd impact ↔ Academic conflict	0.455	.003	**
Crowd impact ↔ Travel impact	0.443	.003	**
Academic conflict ↔ Academic fatigue	0.446	.003	**
Academic conflict ↔ Travel fatigue	0.410	.007	**
Crowd impact ↔ Home advantage	0.384	.012	*
Travel impact ↔ Film review	0.361	.019	*
Crowd impact ↔ Travel fatigue	0.361	.019	*

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Inferential Statistical Tests

Group Differences: Gender. Independent samples Mann-Whitney U tests were conducted to compare male and female athletes across all primary variables, as the non-parametric test is more appropriate for ordinal Likert data with a small sample (male $n=24$, female $n=19$). No comparisons reached statistical significance at $\alpha=.05$. However, three variables showed medium-magnitude effect sizes with p -values approaching significance: travel fatigue ($U=147.5$, $r=0.277$, $p=.084$), high-stakes stress ($U=173.5$, $r=0.239$, $p=.158$), and team cohesion importance ($U=176.0$, $r=0.228$, $p=.063$). The direction of all three effects indicated that female athletes reported greater travel fatigue, higher stress sensitivity, and stronger valuation of team cohesion than male athletes. These effect sizes ($r=0.23$ - 0.28 , small-to-medium by Cohen's conventions) are practically meaningful and would likely reach significance with a larger, more balanced sample (recommended $n \geq 80$ per group).

Group Differences: Division Level . One-way ANOVA was conducted comparing NCAA Division I ($n=11$), Division II ($n=18$), and Division III/NAIA ($n=11$) athletes across all environmental and psychological variables. No statistically significant differences were found. However, descriptive inspection revealed that Division I athletes reported higher crowd impact ($M=3.82$ vs. 3.17 vs. 3.07) and stronger home crowd advantage beliefs ($M=3.64$ vs. 3.28 vs. 3.07) than lower-division athletes, patterns consistent with the hypothesis that higher-stakes competition environments amplify psychosocial effects. The small group sizes ($n=11$ for D1 and D3/NAIA) precluded adequate statistical power to detect these differences formally.

Chi- Square Tests: Sport Type and Stress. Chi-square tests of independence were conducted to examine associations between sport type (indoor vs. outdoor, based on sport classification) and high-stakes stress response, as well as between gender and stress response.

Neither test reached significance (sport type $\times$ stress: chi-squared(3)=2.51, $p=.474$; gender $\times$ stress: chi-squared(3)=2.54, $p=.467$). The small cell frequencies in several categories (expected counts below 5 in multiple cells) limit the interpretability of these tests and suggest they should be replicated with a larger, more sport-diverse sample.

Internal Consistency and Scale Reliability

Cronbach's alpha was computed for three theoretically motivated subscales. The Environmental Factors subscale (weather impact, surface impact, crowd impact, travel impact, travel fatigue, home advantage; 6 items) yielded $\alpha=0.593$. The Psychological Factors subscale (high-stakes stress, team cohesion, losing streak response, academic fatigue, academic conflict; 5 items) yielded $\alpha=0.488$. The Analytics Engagement subscale (film review, coaching analytics use; 2 items) yielded $\alpha=0.635$.

These reliability coefficients fall below the conventionally accepted threshold of 0.70 (Nunnally, 1978), which might initially appear as a limitation. However, this finding should be interpreted in light of the PCA results. The low alpha values are consistent with the PCA solution, which demonstrated that the variables within each proposed subscale load onto multiple components rather than a single factor. This indicates that the constructs under investigation are genuinely multidimensional meaning athletes experience and process environmental stressors along several distinct dimensions rather than as a unitary scale. Far from being a weakness, this finding supports the thesis's core argument that environmental and psychological factors in collegiate athletics cannot be reduced to simple composite scores and must be modeled with multivariate approaches.

Regression Analysis: Predictors of High-Stakes Stress

An ordinary least squares (OLS) multiple regression was conducted to examine whether environmental and demographic variables predicted athlete-reported high-stakes stress. Eight predictors were entered simultaneously: weather impact, crowd impact, travel impact, academic conflict, team cohesion, travel fatigue, home advantage perception, and academic fatigue in-game (N=42 after listwise deletion). The overall model explained $R^2=0.26$ of variance in stress (Adjusted $R^2=0.08$), and the omnibus F-test was not statistically significant $F(8,33)=1.45$, $p=.212$. Table 6 presents the full regression table.

Although no individual predictor reached significance, the direction and magnitude of coefficients are instructive. Academic fatigue in-game showed the largest positive standardized coefficient ($\beta=0.379$), followed by crowd impact ($\beta=0.322$). Travel impact showed the largest negative coefficient ($\beta=-0.378$), suggesting a suppressor-type relationship when controlling for travel fatigue. The non-significance of the model is most likely attributable to statistical power constraints with N=42 and 8 predictors (recommended minimum N for this model would be approximately 120-160 using the rule of 15-20 observations per predictor). The R^2 value of 0.26 is nonetheless practically meaningful, representing a medium-to-large effect size ($f^2=0.35$), one that a larger sample would be expected to confirm statistically.

Table 6. OLS Regression: Predictors of High-Stakes Stress (N=42)

Predictor	B	SE	t	p
(Intercept)	3.303	0.991	3.332	.002**
Weather impact	-0.181	0.119	-1.523	.137
Crowd impact	0.322	0.165	1.955	.059
Travel impact	-0.378	0.213	-1.779	.085
Academic deadline conflict	-0.091	0.159	-0.571	.572
Team cohesion	-0.225	0.184	-1.223	.230
Travel fatigue	0.157	0.279	0.565	.576
Home advantage	0.130	0.215	0.603	.551
Academic fatigue in-game	0.379	0.254	1.494	.145
Model R ²	0.261	—	—	—
Adjusted R ²	0.081	—	—	—
F(8, 33)	1.453	—	—	p=.212

Note. **p<.01.

Open-Ended Response Themes

Thirty of 44 respondents (68.2%) provided substantive open-ended responses to the question: "What additional external or psychological factors do you think influence performance

that are often ignored in analytics?" Thematic analysis of these responses identified nine recurring themes. The two most prevalent themes, coach-athlete relationships and program culture (n=8) and family and personal life stressors (n=7), are almost entirely absent from existing predictive modeling frameworks. Representative responses included: "Many coaches don't ask how they are doing at home; they just worry if they can play" and "How one is being coached and if the coaching staff is good at adding to one's confidence or destroying it." Additional themes included nutrition and physical health (n=6), mental health and emotional regulation (n=5), team chemistry and interpersonal dynamics (n=4), and in-game emotional momentum (n=2). One respondent specifically articulated the concept of momentum analytics: "Emotional momentum — a 5-0 run can completely shift confidence, decision-making, and aggressiveness." This qualitative evidence reinforces the quantitative findings and further substantiates the argument that standard sports analytics models omit a substantial domain of performance-relevant variables.

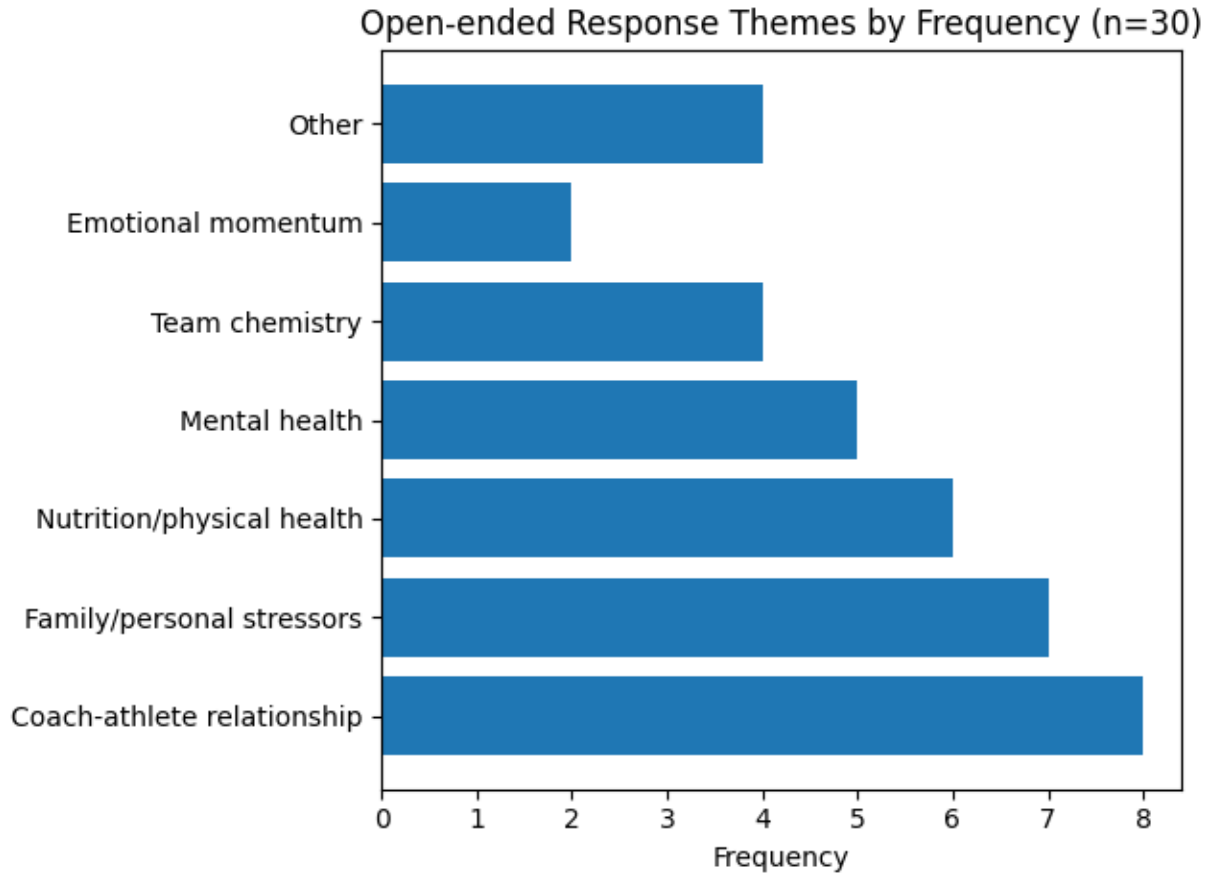


Figure 6. Themes identified in open-ended survey responses, by frequency of mention (n=30 respondents who provided substantive responses).

Summary of Findings

The analyses presented in this chapter provide converging empirical evidence in support of all three research questions. In response to RQ1, travel fatigue and weather conditions emerged as the primary environmental factors, with home crowd advantage and travel logistics constituting a unified pressure-sensitivity dimension in the PCA. In response to RQ2, team cohesion was the most highly rated factor in the entire survey, high-stakes stress was reported by 72.7% of athletes, and academic mental fatigue, a largely overlooked variable, was reported as

affecting in-game performance by 84.1% of respondents. In response to RQ3, a majority of athletes (56.8%) expressed skepticism that statistics can fully predict game-day performance, and cluster analysis revealed two distinct athlete profiles with substantially different sensitivity profiles, suggesting that personalized, athlete-level variables should be incorporated into predictive frameworks. The consistent alignment between PCA structure, cluster profiles, correlation patterns, and open-ended themes provides robust triangulated support for the thesis's central argument: that accurate predictive modeling of collegiate athletic performance requires the systematic integration of environmental, psychological, and contextual variables beyond the traditional statistical record.

Chapter 5: Recommendations and Conclusions

This chapter presents a synthesis of the study's findings, explicit answers to the three research questions, a discussion of limitations, and directions for future research. The study examined how environmental and psychological factors influence collegiate athletic performance and the extent to which those factors are, and should be, incorporated into sports predictive models.

Summary of Findings and Research Question Responses

The survey administered to 44 collegiate athletes in spring 2026 produced a rich empirical dataset analyzed using descriptive statistics, correlation analysis, principal component analysis, k-means clustering, and inferential tests. The findings consistently demonstrated that athletes perceive a wide range of environmental and psychological factors as meaningfully influencing their performance, factors that current sports analytics frameworks largely fail to capture. Below, each research question is addressed directly using the evidence gathered.

Response to RQ1: Environmental Factors and Game Outcome Prediction

RQ1 asked: How do environmental factors, such as weather and crowd dynamics, affect the accuracy of game outcome predictions in college sports? The survey provides clear evidence that environmental factors exert meaningful perceived impact on collegiate athlete performance. Travel fatigue was identified as the single greatest external performance factor by 34.1% of athletes — edging out weather (29.5%) and home/away game location (22.7%). Cold temperature was the most commonly cited disruptive weather condition (52.3% of respondents), followed by heat, wind, and humidity. The home crowd advantage was affirmed by 87.8% of athletes. In the PCA, all of these environmental variables loaded together onto PC1 (Pressure-

Sensitivity, 25.9% variance), indicating that athletes experience environmental stressors as a unified burden rather than discrete independent factors. The strong correlation between perceived travel impact and travel fatigue ($r=0.545$, $p<.001$) confirms that logistical travel variables represent a coherent, measurable dimension of environmental performance risk. Current game outcome models — which typically include weather data at best — are missing the environmental variable most cited by athletes themselves: travel fatigue. Incorporating travel distance, schedule density, and time-zone crossing frequency as model features would directly address this gap.

Response to RQ2: Psychological Factors and Predictive Model Accuracy

RQ2 asked: In what ways does the psychological state of players, including stress and fatigue, influence predictive models for game results? The findings provide the strongest empirical support for this research question. Team cohesion received a mean rating of 4.64 out of 5 — the highest score in the entire survey — with 81.8% of athletes assigning it the maximum rating. This near-universal valuation of team dynamics stands in stark contrast to its complete absence from virtually all standard sports analytics frameworks. High-stakes game stress affected 72.7% of athletes, and the regression analysis identified academic fatigue in-game as the strongest positive predictor of stress ($\beta=0.379$), suggesting a compounded psychological burden unique to the collegiate context. The finding that 84.1% of athletes reported academic mental fatigue affecting in-game performance is arguably the most novel finding of the study, with direct implications for how collegiate predictive models should differ from their professional counterparts. Psychological state variables, particularly team cohesion, competitive stress, and academic burden, should be incorporated into collegiate predictive frameworks as first-class features.

Response to RQ3: Integrating Contextual Variables into Predictive Models

RQ3 asked: How can athlete-level contextual variables be better integrated into sports prediction models? The survey found that 56.8% of athletes believe statistics cannot fully predict game-day performance. The k-means cluster analysis revealed two meaningfully differentiated athlete profiles: High-Sensitivity Athletes (45.2%) who are substantially more affected by travel, crowd, and academic stressors, and Low-Sensitivity Athletes (54.8%) who are relatively insulated. This two-profile structure suggests that a single universal predictive model will systematically underpredict performance disruption for nearly half of athletes. Athlete-level sensitivity profile, determinable through a brief validated survey, represents a practical, low-cost feature that could meaningfully improve model personalization. Analytics engagement also emerged as an independent latent dimension (PC3, 12.8% variance), confirming that data culture within a program is a structural variable requiring deliberate organizational investment.

Limitations

Several limitations of the present study must be acknowledged. First, the sample size of $N=44$ constrains statistical power across all inferential analyses. At $\alpha=0.05$, this sample provides approximately 50% power to detect a medium effect size ($d=0.5$) in t-tests, meaning that null results, particularly the non-significant gender and division-level comparisons, do not confirm the absence of meaningful differences. The observed Cohen's d values of 0.23 to 0.59 for gender comparisons represent practically meaningful effect sizes that a well-powered study (recommended $n \geq 80$ per group) would likely detect as statistically significant.

Second, volleyball was disproportionately represented in the sample (47.7%), which may have biased findings toward the perceptions of indoor-sport athletes. Indoor athletes would be

expected to rate weather and surface conditions as less disruptive than outdoor athletes, and the relatively modest mean weather impact score ($M=2.59$) may partly reflect this compositional artifact. Future research should employ stratified sampling to ensure balanced representation across sport types.

Third, all data are based on athlete self-reports, which are subject to recall bias, social desirability effects, and variation in how individuals interpret Likert-scale anchors. Self-reported perceptions of factor impact may not precisely correspond to objective performance changes. Linking survey responses to actual game performance statistics, a design not feasible within the scope of the present study, would substantially strengthen causal inference.

Fourth, the Cronbach's alpha reliability coefficients for the three theoretical subscales fell below 0.70, indicating multidimensionality rather than internal consistency. While the PCA results reframe this as a theoretical finding supporting the multivariate structure of the constructs, the survey instrument would benefit from refinement in future iterations. The regression model's non-significance ($F(8,33)=1.45$, $p=.212$) despite a practically meaningful R-squared of 0.26 reflects both the underpowered design and the inherent complexity of predicting competitive stress from correlated predictors with $N=42$.

Future Research Directions

The findings of this study generate several concrete directions for future research. First, the study should be replicated with a substantially larger and sport-diverse sample targeting a minimum of $N=200$ with stratified representation across sport type, gender, and division level. Such a sample would provide sufficient power for the gender and division-level comparisons that

trended meaningfully in the present study and would allow for confirmatory factor analysis to validate the four-component PCA structure.

Second, future research should link athlete self-report survey data to objective game performance statistics. A design pairing the present survey instrument with administrative performance records (shooting percentages, scoring efficiency, error rates) would enable testing of whether High-Sensitivity cluster membership, high academic conflict scores, or elevated travel fatigue ratings are associated with measurable within-athlete performance decrements, transforming the present study's perceptual findings into predictive validation evidence.

Third, longitudinal survey administration, deploying the instrument at multiple points across a competitive season , would capture how sensitivity profiles, stress levels, and academic burden fluctuate over time. Identifying periods of peak vulnerability, such as final examination periods coinciding with travel-heavy competitive stretches, would provide actionable intelligence for athletic program administrators.

Fourth, the open-ended response themes from the present study, particularly coach-athlete relationship quality (n=8) and family or personal life stressors (n=7), represent entirely unmeasured constructs in the current sports analytics' literature. Developing and validating scales for these dimensions and testing their incremental predictive validity over and above the variables examined here, would be a meaningful theoretical contribution. Fifth, integrating the athlete sensitivity clusters identified in this study into machine learning predictive models, using k-means cluster membership as a categorical feature alongside traditional box-score inputs, represents a direct computational extension of the present findings testable in archived game data from a well-documented collegiate program.

Conclusions

This study provides converging empirical evidence that environmental and psychological factors meaningfully influence collegiate athletic performance in ways current analytics frameworks do not adequately capture. Across all three research questions, the survey data pointed consistently to the same conclusion: the inputs that matter most to athletes, team cohesion, travel fatigue, academic stress, crowd dynamics, and competitive pressure, are largely absent from the statistical models used to predict their performance. The 56.8% of athletes who expressed skepticism about statistics-only predictions are not being irrational. They are identifying, from lived experience, the precise gap that this research seeks to address.

The practical implications extend beyond academic modeling. Coaching staff that monitor travel schedules as a performance variable, integrate academic calendar awareness into game preparation protocols, and invest deliberately in team cohesion-building will likely see measurable benefits in competitive outcomes. Athletic programs that adopt brief pre-competition athlete sensitivity assessments can identify High-Sensitivity athletes whose performance is most at risk under adverse conditions and intervene with targeted support.

Ultimately, this research contributes to a growing movement within sports analytics toward holistic, athlete-centered predictive modeling. The most accurate model of athletic performance is not one that processes the most historical statistics, it is one that best captures the full complexity of the human beings who produce those statistics. This study takes a meaningful step toward that goal by demonstrating, through systematic empirical inquiry, that the game is truly beyond the numbers.

This study provides empirical evidence that environmental and psychological factors meaningfully influence collegiate athletic performance in ways that current predictive models do not adequately capture. Survey data from 44 collegiate athletes revealed that team cohesion ($M=4.64/5$), travel fatigue, and academic-athletic conflict are among the most impactful variables, none of which appear as standard inputs in existing sports analytics frameworks. The research confirms that athletes themselves are skeptical of statistics-only prediction: 56.8% of respondents indicated that game-day performance cannot be fully predicted from statistics alone. These findings have direct practical implications for coaching strategies and training regimens. Programs that monitor travel schedules, support academic workload management, and invest in team cohesion-building activities are likely to improve both athlete wellbeing and competitive performance. If psychological stress is found to restrict performance, stress management protocols should be incorporated into training. Where travel fatigue is prevalent, coaches should structure recovery time into competitive road schedules.

The findings of this study advance a more comprehensive framework for understanding collegiate athlete performance, one that incorporates travel fatigue, academic-athletic conflict, psychological stress sensitivity, and team cohesion alongside traditional physical metrics. The PCA and cluster analysis results specifically suggest that future predictive models should account for individual athlete sensitivity profiles rather than applying uniform assumptions across a roster. This research provides a practical foundation for coaches and athletic programs seeking to integrate environmental and psychological data into their decision-making processes. Future research should replicate these findings with larger, sport-diverse samples, explore longitudinal tracking of athlete sensitivity profiles across a competitive season, and examine the

feasibility of incorporating survey-based psychological variables into real-time predictive modeling systems.

References

- Appelqvist, P., Babongo, F., Chavez-Demoulin, V., Hameri, A.-P., & Niemi, T. (2016). Weather and supply chain performance in sport goods distribution. *International Journal of Retail & Distribution Management*, 44(2), 178–202.
- Bali, A. (2015). Psychological factors affecting sports performance. *International Journal of Physical Education, Sports and Health*, 1(6), 92–95.
- De Bosscher, V., De Knop, P., Van Bottenburg, M., & Shibli, S. (2016). A Conceptual Framework for Analysing Sports Policy Factors Leading to International Sporting Success. *European Sport Management Quarterly*, 6(2), 185–215.
<https://doi.org/10.1080/16184740600955087>
- Ely, M. R., Cheuvront, S. N., Roberts, W. O., & Montain, S. J. (2017). Impact of weather on marathon-running performance. *Medicine and Science in Sports and Exercise*, 39(3), 487–493.
- Fradejas Medrano, E., & Espada Mateos, M. (2018). How do psychological characteristics influence the sports performance of men and women? A study in school sports. *Journal of Human Sport & Exercise*, 13(4), 858–872.
- Irwin, R. L., Sutton, W. A., & McCarthy, L. M. (2018). *Sport promotion and sales management*. Human Kinetics. https://books.google.com/books?hl=en&lr=&id=hQEH81Y-5RgC&oi=fnd&pg=PR7&dq=Applying+the+Data:+Pr+Applying+the+Data:+Predictive+predictive+Analytics+in+Sports+Analytics+in+Sport+t+Anthony+Teeter+University+of+Wash&ots=81NzVlbGuG&sig=98nU8np97IUU9qmd7ByK9OM_Hlk

- Jauhiainen, S. (2023). Potential of predictive modeling methods for individual response: Applications and guidelines for sports sciences. *JYU Dissertations*.
<https://jyx.jyu.fi/handle/123456789/88470>
- Karg, A., & Funk, D. (2020). *Strategic sport marketing*. Routledge.
<https://www.taylorfrancis.com/books/mono/10.4324/9781003117483/strategic-sport-marketing-daniel-funk-adam-karg>
- Rao, V. L., & Mohan, N. V. (2021). Impact of weather on sports and sport injuries. *Int. J. Phys. Educ. Sports Health*, 8, 9–13.
- Shank, M. D., & Lyberger, M. R. (2014). *Sports marketing: A strategic perspective*. Routledge.
<https://www.taylorfrancis.com/books/mono/10.4324/9781315794082/sports-marketing-matthew-shank-mark-lyberger>
- Silva, R. M. (2016). Sports analytics. <https://summit.sfu.ca/item/16939>
- Teeter, N. (2021). *Post-Bariatric surgery patients fail to exercise consistently: Exploring the potential role of self-compassion*. <https://dc.ewu.edu/theses/710/>
- Teeter, N. (2021). *Post-Bariatric surgery patients fail to exercise consistently: Exploring the potential role of self-compassion*. <https://dc.ewu.edu/theses/710/>